Supplementary materials

This file accompanies the main paper. It holds the calibration check, the cue lists, the statement links, the non-electrocardiogram controls, the numeric-card and identifier experiments, the incomplete open-model parses, the earlier samples, and the discussion of those checks. Each section names the table or result in the main paper that it complements.

1 Methods S1. Full calibration diagnostics for keyword and TF-IDF retrieval

This section complements Table 1 of the main paper, which lists the criteria library used for retrieval.

On a 300-patient calibration diagnostic, keyword retrieval produced 12 distinct top-three sets and 15 distinct cards, with a distractor fraction of 0.639. TF–IDF produced 14 sets and 16 cards, with a distractor fraction of 0.551. The sets differ by superclass, which is the property the retired query lacked. Conduction disturbance tends to retrieve a narrow-QRS card, a pediatric QRS card, and the wide-QRS card. ST/T change tends to retrieve long QTc, ST depression, and an ST-elevation card. Normal tends to retrieve cards about capture beats, sinus rate variation, and an upright T wave. The pediatric QRS card and the capture-beat card are examples of untestable items: they are textually close to the query and have no check. A model that marks them as applying creates a release-score event that the withdrawal rule later removes by design.

2 Methods S2. Full affirmation and refusal lists

This section complements the cue-list rule behind the exploratory diagnosis rates in Table 7 of the main paper.

A sentence counts as affirmed when it contains a positive cue such as “possible”, “probable”, “suggested”, “indicated”, “consistent with”, “are present”, “is present”, “shows a”, or “most likely”. It does not count as affirmed when a refusal cue is present, including “no evidence”, “no clear evidence”, “not confirmed”, “not met”, “does not reach”, “below the threshold”, and “no pathologic”, even if a positive cue is also present. Writing the superclass code alone, which Qwen3-8B often does, is not an affirmation. The affirmation rule is a text rule, applied after generation, and identical for every arm. A criterion claim counts as applied when the model sets the applies flag to true, whether or not the sentence would also have matched the affirmation list.

3 Methods S3. Statement links for the original eight checks

This section complements Table 5 of the main paper, which reports statement mismatch for the original eight checks.

The links were fixed before scoring: wide QRS to the bundle-branch and pre-excitation codes; long PR to first-degree block and a long-PR code; left axis to left anterior fascicular block; Sokolow–Lyon to the left-ventricular-hypertrophy and high-voltage codes; long QTc to the long-QT code; ST depression to the ST-depression code; a long Q wave to the Q-wave code; and tachycardia to the sinus and supraventricular tachycardia codes. Precision of applied linked checks is the fraction of those applications for which the linked code is present. Mean unstated findings are the average, over records, of the number of linked codes present in the reference and absent from the note. Distractors and the six opposing checks do not enter that average. Unknown card identifiers count toward the release score only.

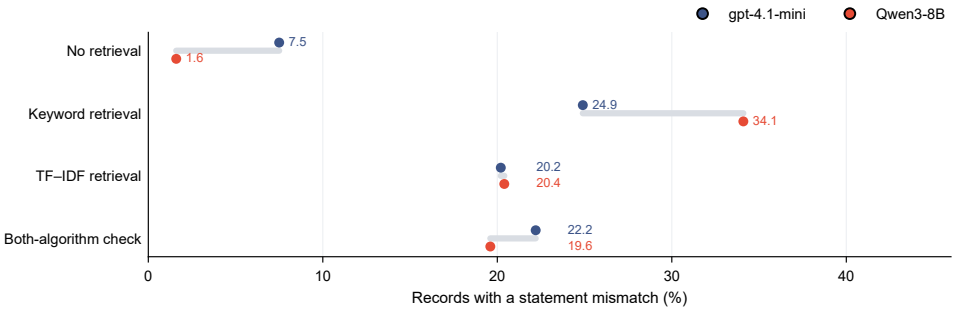


Fig. S1 Statement mismatch for the original eight checks. This figure complements Table 5 of the main paper.

4 Methods S4. Non-electrocardiogram recordings, noise, and the waveform network

This section complements the waveform and specificity-control results in the main paper.

The same note rule was applied, with gpt-4.1-mini only, to two recordings that are not electrocardiograms. These checks ask whether retrieval still creates unsupported criteria and whether the measurement edit still removes them. They are not a second clinical claim and are not evidence about PTB-XL; they bound the sentence "retrieved cards cause unsupported applications, and a measurement edit removes them" by asking whether that sentence survives a change of recording.

The resting electroencephalogram set is OpenNeuro ds004504, 88 eyes-closed recordings from 36 participants with Alzheimer's disease, 23 with frontotemporal dementia, and 29 controls. The recordings are stored inside MATLAB 5 .set files. Six spectral cards have a numeric check, including a posterior peak below 8 Hz, a theta-to-alpha ratio of at least 1, and relative theta of at least 0.25, together with opposing cards and distractors that have no check. The posterior peak is the argmax, inside 6–14 Hz, of the power spectrum divided by a 1/f floor, after a 1 Hz high-pass. A logistic

regression, fit inside each leave-one-out fold, supplies the group probabilities. Six participants were a pilot, seed 21; the scored set is the other 82. An affirmed group with probability below 0.5, or an applied criterion the spectrum does not support, counts as an unsupported record. The edit withdraws an applied criterion when the check is missing or the predicate is false. Group sentences are not rewritten.

The sleep check uses eight nights from the Sleep-EDF cassette study on PhysioNet. Four nights train a balanced logistic regression on six epoch features; four nights are held out. The hypnogram suffix is not the same on every night. The signal header of a polysomnogram stores fields across channels. It does not store one record per channel. Measurements on each 30-second epoch were frozen on the training night before any note was requested. A single maximum 2-second peak-to-peak in a 0.5–2 Hz band marked most of wake as positive, because blinks survive that filter, so that definition was not used. The slow-wave measurement is instead the fraction of non-overlapping 1-second bins whose 0.5–2 Hz peak-to-peak amplitude on Fpz–Cz is at least 75 μV. The spindle measurement is the longest run of an 11–16 Hz envelope at or above 5 μV. Relative alpha is power from 8–13 Hz over power from 1–30 Hz on Pz–Oz. Cards state slow-wave activity on at least 20% of the epoch, a spindle of at least 0.5 s, and relative alpha of at least 0.15, plus opposing cards and distractors with no check. The alpha threshold of 0.15 was chosen on the training night because a threshold of 0.25 was almost empty in wake; it was not retuned after the notes. The scored sample is 12 epochs from each of wake, N1, N2, N3, and REM, seed 7, after one pilot epoch per stage. The case rate counts an affirmed stage whose probability is below 0.5, or an applied criterion the measurements do not support. Stage sentences are not rewritten. A second count uses only the criterion term, so a failure that lives entirely in the stage sentence is visible.

Synthetic noise, with gates set at the 99th percentile of a calibration sample, and the MIT–BIH Noise Stress Test records were mixed into a frozen sample of 100 electrocardiograms. The mixes describe predicate stability; they were not passed back through the language model. A four-head convolutional network was trained to detect wide QRS, long PR, left axis, and tachycardia from 12 leads at 100 Hz for 10 seconds, taken from the 500 Hz files by keeping every fifth sample. Two hundred MIMIC-IV-ECG studies were used only to test that network. MIMIC machine measurements can support part of the library and cannot support Sokolow–Lyon voltage, ST amplitude in II, Q duration, or sex-specific QTc, so the eight-check note comparison was not moved there.

5 Methods S5. Numeric cards only

This section complements Table 6 of the main paper, which reports retrieval from the 14 cards that have a numeric check.

The second generation asked gpt-4.1-mini for keyword and TF–IDF notes after the 33 cards with no check had been removed. All 1,083 patients parsed. Keyword prompts contained 1,431 cards in 15 distinct sets. The mean prompt therefore contained fewer than three cards, because some superclass queries overlap only one or two of the 14 numeric cards, and none of those slots was a distractor. The retriever still varied: fifteen

distinct sets on a 14-card library is a different experiment from the retired query that returned one set for almost every record. The no-retrieval column is the original note.

The release score fell from 46.8% on the full library to 13.8%. That is the quantitative content of the claim that the full-library drop was largely an artifact of untestable cards. It is not a claim that the artifact was the whole story. The mean note applied 0.386 criteria that 12SL supported and 0.024 that it rejected. Another 0.118 applied criteria had no usable card identifier. Of those 128 applications, 113 left the identifier empty and 15 named an identifier outside the 14 cards. Every one of the 128 was absent from the cards actually shown in that prompt. The model was not applying a distractor the retriever had retrieved, because the retriever had none to retrieve; it was emitting a criterion claim that the prompt's identifier rule did not constrain. The 12SL withdrawal sets the release score to zero by construction, including those identifier failures, because a missing check is withdrawn.

Uni-G rejection moved from 7.9% to 5.5%. The difference was 0.024 (interval 0.016 to 0.033), from 26 records against 0. The agreement rule leaves 0.330 criteria per note, each accepted by both algorithms, and sets the Uni-G rate to zero because Uni-G is then an input. Compared with the full library, the numeric-only prompts state more supported criteria per note, 0.386 against 0.294 before the agreement rule and 0.330 against 0.247 after it. Removing untestable cards from the top three leaves more of the top three available for predicates that can hold.

The expert-label rate was 9.8%, against 7.5% with no retrieval. The difference was 0.023 (interval 0.006 to 0.041), from 59 records against 34, and the withdrawal left it unchanged. A library that contains only testable cards does not remove the diagnosis effect of showing cards to the model. Statement mismatch was 25.6% after keyword retrieval, 24.7% after the 12SL withdrawal, and 22.3% after the agreement rule, close to the full-library figures of 24.9%, 24.6%, and 22.2%. The linked code was present for 34.0% of 338 keyword applications and for 39.3% of the 275 applications both algorithms kept. Precision is higher than the 23.9% and 27.5% of the full library, which is expected once fewer unlinked and untestable cards compete for the applied set, and it remains well below half. TF–IDF retrieval on the numeric library gave a release score of 16.0%, a Uni-G rejection rate of 8.8%, and an expert-label rate of 11.4%.

6 Methods S6. Matched instructions and constrained identifiers

This section complements the identical-schema comparison in Table 6 of the main paper and the different-schema comparison in Table 8.

The matched generation removes the ban on criterion claims. All 1,083 gpt-4.1-mini patients parsed. Without cards, the model applied 0.174 criteria per note. Every one of them lacked a numeric check, so the release score was 17.4% and the second-algorithm rejection rate was zero. The expert-label rate was 7.6%, close to the 7.5% of the original prompt that forbade criteria.

With the keyword cards under the same instructions, the release score was 31.1%. The paired difference from the no-card arm was 0.138 (interval 0.106 to 0.169), from 236 records against 87. The mean note applied 0.299 criteria that 12SL supported,

0.018 that it rejected, and 0.294 with no check. The expert-label rate was 10.6%. The difference from the no-card arm was 0.030 (interval 0.015 to 0.047), from 57 records against 24. The 12SL edit left that expert-label rate unchanged and set the release score to zero by withdrawing 318 unchecked applications and 20 false predicates. Uni-G rejected 19 of those 20 predicates and supported 1. The Uni-G record rate moved from 5.9% to 4.3%. Allowing criterion claims in both arms therefore raises the no-card release score from the zero of the original prompt to 17.4%. Keyword retrieval remains higher, and the expert-label gap remains.

The diagnosis comparison in Table 8 of the main paper is a later pair, with one system instruction for both arms: "Return one JSON object with a claims list of at most 8 claims. Use only the supplied measurement numbers. Do not invent values. A diagnosis claim may use only a label whose probability is at least 0.5. Include one limit claim stating that this is not a clinical diagnosis. Allowed types are diagnosis, measurement, criterion, and limit. Emit a criterion claim only when retrieved_cards is present. Then card_id must be one of those ids and applies is true or false." The user message carries the measurements. Only the constrained arm adds the retrieved cards. The schema admits a criterion claim only in that arm, and the identifier must be one of the retrieved ids. Those cards are the 14 numeric entries of the library named in the code-availability statement. Micro-F1 and hypertrophy-recall differences use 10,000 paired patient resamples with seed 2, and each resample recomputes the pooled counts. The hypertrophy interval uses the 107 patients whose stored code includes hypertrophy.

The schema generation used the 14 numeric cards and the original keyword prompt. All 1,083 notes parsed. Of 469 applied criteria, none had an empty identifier and none used an identifier outside the retrieved set. The release score was 3.0%, against 13.8% when the identifier was free text. The mean note applied 0.397 criteria that 12SL supported and 0.036 that it rejected, and none without a check. The 12SL edit withdrew 39 applications. All 39 had a false 12SL predicate, and Uni-G rejected all 39. The Uni-G record rate moved from 8.7% to 5.7% (difference 0.030, interval 0.019 to 0.040, from 32 records against 0). The expert-label rate was 6.0% before and after the edit. Against the original no-retrieval rate of 7.5%, the difference was -0.015 (interval -0.032 to 0.003), from 40 records against 56. Statement mismatch was 25.4% on the keyword note and 20.8% after the agreement rule. The linked code was present for 38.6% of the 277 applications both algorithms kept. Forcing a valid identifier removes the formatting errors that had inflated the numeric-library release score. It does not produce a clear gain on the stored diagnosis.

7 Methods S7. Open models with incomplete JSON

This section complements the model-completion results in the main paper and the DeepSeek row of Table 7.

Llama 3.1 8B Instruct wrote a file for all 1,083 patients, but the notes were usually code or truncated JSON. Parseable notes numbered 496 without retrieval, 51 with keyword retrieval, and 50 with TF–IDF retrieval. Only four patients had all three notes parseable. No rate from those four patients is reported. The token limit of 1,200

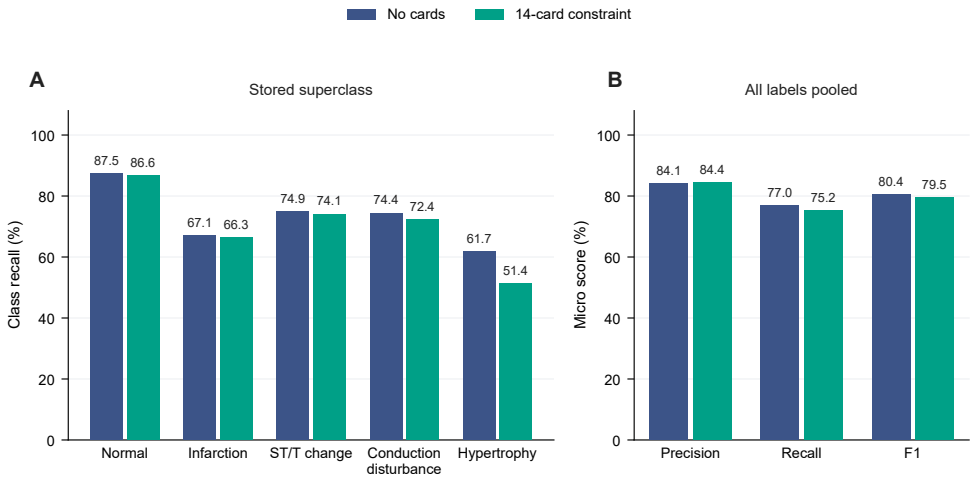


Fig. S2 Different-schema diagnosis agreement for gpt-4.1-mini. This figure complements Table 8 of the main paper.

new tokens, which was sufficient for Qwen3-8B once thinking was disabled, was not sufficient for a model that spent the budget on code. The failure is a generation failure; it is not evidence about criteria.

DeepSeek-R1-Distill-Llama-8B also wrote all 1,083 patients. A long reasoning trace often consumed the token limit before the JSON object closed. An earlier parse had 394 notes without retrieval, 537 with keyword retrieval, and 534 with TF–IDF retrieval, and the paired score below uses the 116 patients with all three notes in that parse. The completed file has 403, 545, and 539 parseable notes and 112 patients with all three. Table 7 of the main paper uses the 112. The combined rates in this section were not recomputed on them. On the earlier 116 the combined rate was 0 without retrieval, 0.388 with keyword retrieval, 0.534 with TF–IDF retrieval, and 0.009 after either measurement check. The keyword-minus-check difference was 0.379 (interval 0.293 to 0.466), from 44 records against 0. Statement mismatch moved from 0.284 with keyword retrieval to 0.172 after the agreement rule, and mean unstated findings stayed at 0.293. This is the same direction as gpt-4.1-mini and Qwen3-8B on the notes that finished as JSON, but it is a selected subset of patients for whom the trace happened to leave room for the object. It is not a replication on the full test fold, and the three-score decomposition reported for the other two models is not repeated on these 116 notes.

8 Methods S8. Second-algorithm score

This section complements the release score, the Uni-G score, and the expert-label score used in Tables 3 and 4 of the main paper.

The second-algorithm score flags a record when Uni-G rejects an applied criterion. For the 12SL edit, Uni-G was not an input, so a change on this score is not guaranteed; for the agreement edit, the score is zero by construction. Withdrawals are counted one

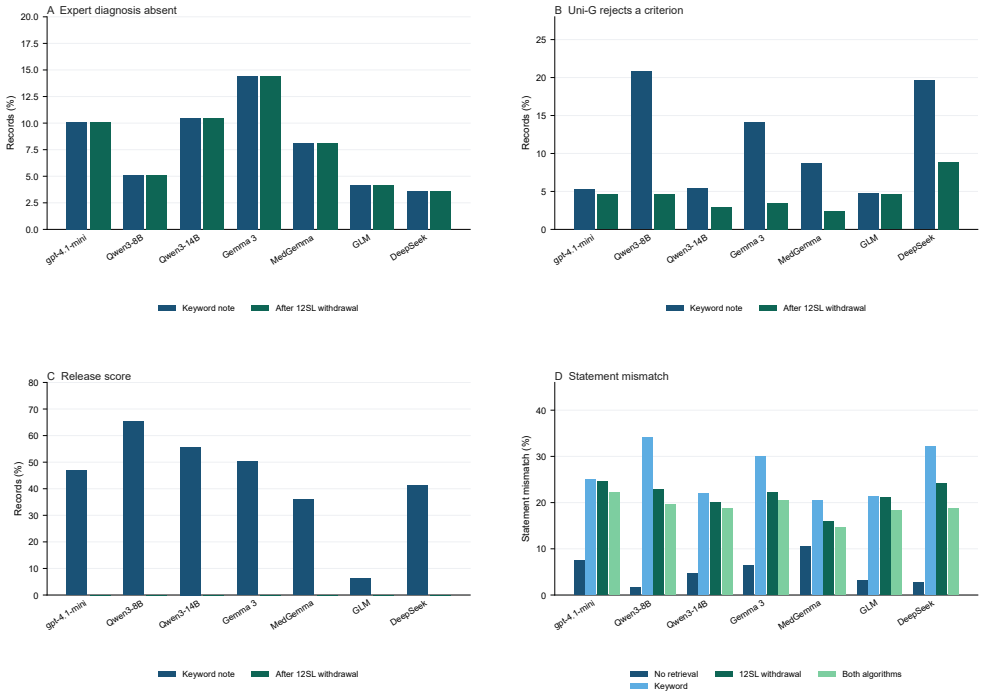


Fig. S3 Keyword notes and the same notes after 12SL withdrawal, across the saved models. The release score after withdrawal is zero, so that bar is absent. DeepSeek is the selected paired subset. This figure complements Tables 3 and 7 of the main paper.

application at a time in three bins: the card had no numeric check; the 12SL predicate was false; and, among the false predicates, Uni-G rejected, supported, or lacked the measurement. The expert-label score flags a record when an affirmed superclass is absent from the PTB-XL diagnostic indicator. The rule does not edit diagnosis text, so this score is identical before and after withdrawal. A separate statement score, limited to the original eight checks, flags an application whose linked cardiologist code is absent or a linked code no claim states.

9 Methods S9. Prompts

This section records the instructions used to generate the notes. Open-model electrocardiogram notes used the same electrocardiogram instructions as gpt-4.1-mini.

The original electrocardiogram instruction, without retrieved cards, was: "Return one JSON object with a claims list of at most 8 claims. Each claim has type, text, and optional label, name, value, card_id, applies. Use only the supplied measurement numbers. Do not invent values. A diagnosis claim may use only a label whose probability is at least 0.5. Include one limit claim stating that this is not a clinical diagnosis. Allowed types: diagnosis, measurement, limit. Do not cite external criteria." The user message was the measurement record. With retrieved cards, the allowed types also included criterion, and the instruction added: "Use the retrieved cards to explain the

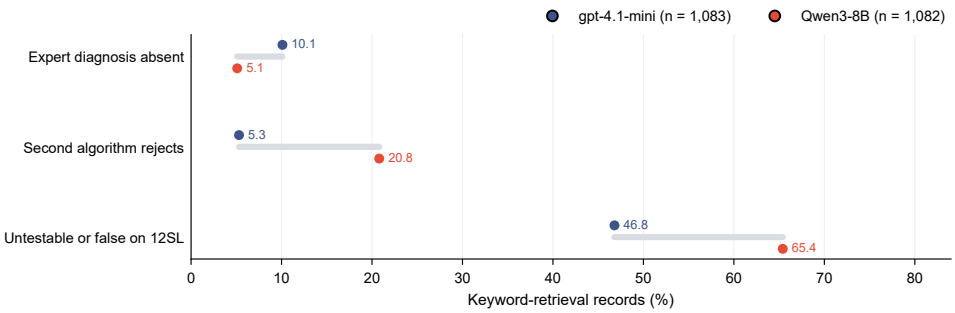


Fig. S4 Expert-label rate, Uni-G rejection, and release score before and after the 12SL withdrawal. This figure complements Table 3 of the main paper.

ECG. A criterion claim must set card_id to one of the retrieved ids and applies to true or false, indicating whether that criterion holds for this patient." The user message then also carried the retrieved card identifiers and text. This is the instruction behind Tables 3, 4, and 5 of the main paper, and behind the 14-card block of Table 6.

The identical-schema instruction was the same for both arms: "Return one JSON object with a claims list of at most 8 claims. Each claim has type, text, and optional label, name, value, card_id, applies. Use only the supplied measurement numbers. Do not invent values. A diagnosis claim may use only a label whose probability is at least 0.5. Include one limit claim stating that this is not a clinical diagnosis. Allowed types: diagnosis, measurement, criterion, limit. A criterion claim sets card_id and applies to true or false." Only the keyword arm added retrieved cards to the user message. This is the instruction behind the identical-schema block of Table 6 of the main paper.

The different-schema instruction is the one quoted in Methods S6. The schema admits a criterion claim only when retrieved cards are present, and the identifier must be one of those identifiers. This is the instruction behind Table 8 of the main paper. The identifier-constrained run on the 14 numeric cards used the original keyword instruction, with the schema limiting card_id to the retrieved identifiers.

The electroencephalogram instruction used the same claim format. A diagnosis label had to be A, F, or C. The instruction said to affirm a diagnosis only when its probability was at least 0.5, and then the text had to say that it was present. The limit claim stated that this was not a clinical diagnosis. Without cards, external criteria were forbidden. With cards, a criterion claim had to set card_id to one of the retrieved identifiers and applies to true or false.

The sleep instruction used the same claim format. A stage label had to be W, N1, N2, N3, or R. The instruction said to affirm a stage only when its probability was at least 0.5, and then the text had to say that it was present. The limit claim stated that this was not a clinical sleep stage. Without cards, external criteria were forbidden. With cards, a criterion claim had to set card_id to a retrieved identifier and applies to true or false.

10 Results S1. Agreement of the measurements, noise, and the waveform network

These checks complement the waveform-network results in the main paper. The transfer numbers are repeated in Table S2.

On the 500 confirmatory electrocardiograms, 12SL and Uni-G agreed closely on the raw measurements and still disagreed often enough that one algorithm should not audit the other by assumption. Median absolute differences were 4 ms for QRS duration (correlation 0.892), 4 ms for PR (0.901, on 472 recordings with both values), 6 ms for QTc (0.873), and 2° for axis (0.903). Heart-rate correlation was 0.999. The median absolute ST difference in lead II was 0.015 mV (correlation 0.784), and the S-wave amplitude in V1 had correlation 0.974. Criterion agreement was 0.982 for wide QRS, 0.964 for long PR, 0.982 for left axis, 0.994 for Sokolow–Lyon, 0.914 for long QTc, 0.992 for ST depression, 0.906 for a long Q wave, and 1.0 for tachycardia. The ST kappa was about 0.50 because the finding is uncommon: high raw agreement and modest kappa are the same fact when almost every tracing is negative. ECGDeli did not match these scales. The median absolute QRS difference was 66 ms (correlation 0.642), the PR difference was 102 ms (correlation 0.612), and the correlation for the S wave in V1 was negative. ECGDeli predicates were withheld. A third algorithm that is not on the same scale is not a second opinion; it is a different measurement.

Cross-scoring the 418 corrected confirmatory notes under the older combined definition showed the practical consequence. Retrieval was unsupported on 0.172 of records against 12SL and on 0.215 against Uni-G. A check computed from 12SL and judged by Uni-G was unsupported on 0.187, a difference of 0.029 from retrieval (interval 0 to 0.060). The reverse direction, a Uni-G check judged by 12SL, separated more cleanly. A quality split of 82 poor recordings and 336 cleaner recordings did not reverse the pattern. Those 418 notes are the reason the agreement mode exists. On the frozen 1,083-patient sample the two release modes then coincided on the older combined rate, for the reason already given: almost every gpt-4.1-mini withdrawal was a card with no check, and both modes withdraw such a card.

Synthetic noise on 100 evaluation recordings, with gates set at the 99th percentile of a calibration sample, showed a narrow condition under which a spectral gate helps. Clean agreement with 12SL was 0.89 for QRS, 0.88 for axis, 0.96 for Sokolow–Lyon, 0.97 for ST depression, 0.86 for Q duration, and 1.0 for heart rate. PR and QTc were withheld from that particular comparison. A 1.0 mV baseline wander still produced a false criterion release near 6.5% after the gate, because the gate does not catch slow drift. Muscle artifact at 0.4 mV and powerline interference at 0.5 mV were largely withheld. They were not accepted as clean measurements. Missing leads did not create false releases, but they did cause naive predicates to call true criteria absent, up to one third of criteria when leads I and aVF were removed together.

Recorded noise from the Noise Stress Test Database, mixed at RMS ratios 0.1, 0.25, and 0.5 onto the same 100 electrocardiograms, kept false releases below 1% at the lower two ratios and reached 2.5% for electrode motion at ratio 0.5. The spectral gate withheld about 3.5% of criteria and did not remove these releases. At a muscle-artifact ratio of 0.5, 11.7% of criteria that were truly present on the clean tracing were

called absent. The noise-file gain field is zero in the source header and was treated as unity. These mixes describe predicate stability under added recorded noise; they were not passed back through the language model, so they do not estimate how often a note would change.

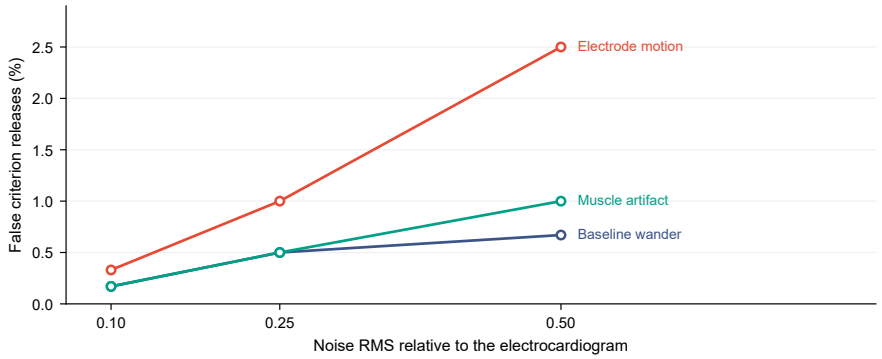


Fig. S5 False criterion releases when Noise Stress Test recordings are added to 100 frozen electrocardiograms. The spectral gate does not remove these releases.

A four-head convolutional network, trained on 15,700 recordings and calibrated on 1,972, reached a calibration macro AUROC of 0.990 and a test macro AUROC of 0.981 on 1,991 recordings for wide QRS, long PR, left axis, and tachycardia. On 200 MIMIC-IV-ECG studies the same heads had AUROC 0.962, 0.957, 0.943, and 0.998, with AUPRC 0.955, 0.592, 0.803, and 0.989 at prevalences 0.125, 0.085, 0.115, and 0.145. Long PR is the predicate whose precision does not travel as well as its discrimination, which is the usual pattern when the finding is uncommon. This is a transfer of four predicates, not of the note protocol. MIMIC-IV-ECG machine measurements, one recording per subject, can support only part of the library. QRS duration at least 120 ms occurred in 9.57% of 161,116 subjects with a known value, against 7.73% of 1,876 PTB-XL test patients. PR at least 200 ms occurred in 8.01% where the interval was present, against 10.02%. Axis at or below $-30°$ occurred in 11.0%, against 10.0%. Heart rate at least 100 occurred in 12.4%, against 6.56%. Sokolow–Lyon voltage, ST amplitude in II, Q duration, and sex-specific QTc are not in that global file.

11 Results S2. Earlier samples and the frozen comparison

These earlier samples precede the frozen comparison in Tables 3, 4, and 5 of the main paper.

The first paired generation on development and test samples scored a broad unsupported rate that counted measurement mismatches and both directions of criterion disagreement. On 129 development records the broad unsupported rate was 0.240. On 180 scored test records it was 0.256 without retrieval and 0.367 with retrieval. A

classifier trained to predict that harm had AUROC 0.610 and AUPRC 0.318, so confidence did not identify the records retrieval would hurt. Literature sentences in that audit had precision 1.0 against the cards that were shown, which is a check that the quoted criterion existed, not that it held for the patient. The audit attempted 200 test electrocardiograms from 193 patients, scored 180, and lost 20 to JSON failures. Under the primary definition used at that time, unsupported rates were 0.050 without retrieval, 0.172 with retrieval, and 0.050 for the circular measurement check. The broad rates on the same notes were 0.261 and 0.367. The retrieved arm contained 48 primary unsupported claims: 35 diagnoses without a supporting probability and 13 criteria applied when the predicate was false. Criterion disagreements numbered 52, of which 39 were true predicates the note did not apply. Per-claim primary unsupported rates were 0.57% without retrieval and 2.3% with retrieval. Record rates and claim rates answer different questions, and the record rate is the one used throughout the frozen comparison because a single applied distractor makes the note carry an unsupported criterion.

A three-policy comparison then appended code-written criterion sentences to a copy of the unretrieved note. That comparison matches the unretrieved note by construction on every claim the copy preserved. It is a circular check. The circular file reproduces those primary rates on the 180 test notes and, on the 129 development notes, gives 0.062, 0.186, and 0.062. Supported diagnoses were retained at rate 1.0 because the check copied them. True criteria were applied at 0.797 by the retrieved note and at 1.0 by the check, again because the check wrote the predicate in code. Those equalities are properties of the copy, and they are why that file is not cited as evidence for the rule.

The confirmatory generation attempted 500 test electrocardiograms and scored 418, with 82 truncated by the 900-token limit. The circular primary rates were 0.036, 0.172, and 0.036. The difference of 0.136 (interval 0.103 to 0.170) came from 57 records that were unsupported only on the retrieved arm. If the 82 missing notes were equal on both arms, the gap on 500 would be 57/500. Bounds that treat every failure as clean, or every failure as unsupported, move the retrieved-arm rate from 0.144 to 0.308. After the refusal list was extended and the same 418 notes were rescored with the corrected edit, the combined rate was 0.036 without retrieval, 0.112 with retrieval, and 0.045 after either check. The statement endpoint was 0.079, 0.376, 0.311, and 0.268. Mean unstated findings stayed near 0.30. Of 57 diagnosis claims sitting under probability 0.5, 22 were affirmed and 35 were refusals, which is why the affirmation list changes the rate. The corrected gap on this smaller sample is about seven points on the statement endpoint between retrieval and the agreement rule, with all 45 discordant records on the retrieved side. The 13-point circular gap is not the frozen result. The frozen result is the three-score decomposition on 1,083 patients, and the numeric-only generation on the same patients.

12 Results S3. Specificity controls outside the electrocardiogram

The table below complements the electroencephalogram and sleep specificity sentence in the main paper.

Table S1 The measurement rule outside the electrocardiogram. The case rate counts an affirmed label below probability 0.5 or an applied criterion the measurements do not support. The sleep criterion row counts only the criterion term.

Check	n	No retrieval	Keyword	After the edit
Resting EEG, case rate	82	0.085	0.024	0.024
Sleep-EDF, case rate	60	0.200	0.650	0.650
Sleep-EDF, criteria only	60	0	0.050	0

On the 82 electroencephalogram subjects, keyword retrieval lowered the unsupported-record rate from 0.085 to 0.024. The measurement edit left it at 0.024. The difference between keyword retrieval and the check was 0. The true group was left unstated in about two thirds of notes under every arm. Retrieval did not create a set of unsupported criteria for the rule to remove. The spectral cards were available. The distractors were available. The model largely declined to apply them. A rule that withdraws unsupported applications has nothing to do when the applications are not made. The leave-one-out group model is a weak probability source. Its accuracy was 0.489 against a majority rate near 0.41. An affirmed group with probability below 0.5 is an easy event to produce. On this sample, it was not the event the cards produced.

On 60 Sleep-EDF epochs the unsupported-record rate rose from 0.200 without retrieval to 0.650 with keyword retrieval and 0.667 with TF–IDF retrieval. The measurement edit left the keyword arm at 0.650. Decomposing that case rate shows why, and the decomposition is the same discipline used for the electrocardiogram. Applied criteria the measurements reject occurred in none of the notes without retrieval, in 0.050 of keyword notes, and in none after the edit. Affirmed stages with probability below 0.5 occurred in 0.200 of notes without retrieval and in 0.650 with the cards. The edit does not change stage sentences, so the case rate does not move. The true stage was left unstated in 0.467 of notes without cards and in 0.933 with keyword cards. Showing the cards made the model state a stage, and the stage it stated was often one the probability model had not supported.

The stage model itself is not an innocent backdrop. On all held-out epochs it matched the hypnogram on 0.577 of epochs. Median probability for true N2 was 0.970 and for true N3 was 0.913, while wake, N1, and REM were poorly separated. Restricting the score to the 32 epochs whose true stage had probability at least 0.5 removes the excuse that the cards were merely echoing a bad probability. On those 32 epochs, unsupported stages were 0.062 without cards and 0.875 with them, while unsupported criteria were 0.031 and then 0 after the edit. The cards caused stage errors. The measurement rule, which is not allowed to edit stage sentences, left the errors in place.

Neither the electroencephalogram check nor the sleep check is evidence that the electrocardiogram edit fails on PTB-XL. Both are evidence that the sentence "retrieval creates unsupported criteria and the edit removes them" is a statement about this library and this note type, not a statement about retrieved clinical text in general.

13 Results S8. What the second algorithm confirms

These counts complement the independent-verification results in Table 4 of the main paper.

The second-algorithm score is the one the 12SL edit does not define. On gpt-4.1-mini it was 5.3% before the edit and 4.6% after. The rule withdrew 647 applications. Of those, 639 had no numeric check, so Uni-G had no predicate to evaluate. The value was missing. It was not a confirmed disagreement. Eight failed 12SL. Uni-G rejected seven of the eight. Almost no numeric criterion that 12SL rejected was applied by the closed model.

On gpt-4.1-mini, Uni-G supported 269 applied criteria and the edit kept 268; on Qwen3-8B, Uni-G supported 282 and the edit kept 274. The edit keeps nearly every criterion the second algorithm supports.

Qwen3-8B is the replication in which the predicate itself fails often. Keyword retrieval put the release score at 65.4%. The rule withdrew 1,091 applications: 880 with no check and 211 with a false 12SL predicate; Uni-G rejected 203 of the 211. The Uni-G record rate fell from 20.8% to 4.7%. Because the edit did not read Uni-G, that drop is not by construction: it is the empirical statement that criteria Qwen applied against 12SL were usually also unacceptable to the second algorithm. The expert-label rate was 1.6% without retrieval and 5.1% with it, unchanged by withdrawal. After the agreement rule, 0.253 criteria per note remained, each accepted by both algorithms.

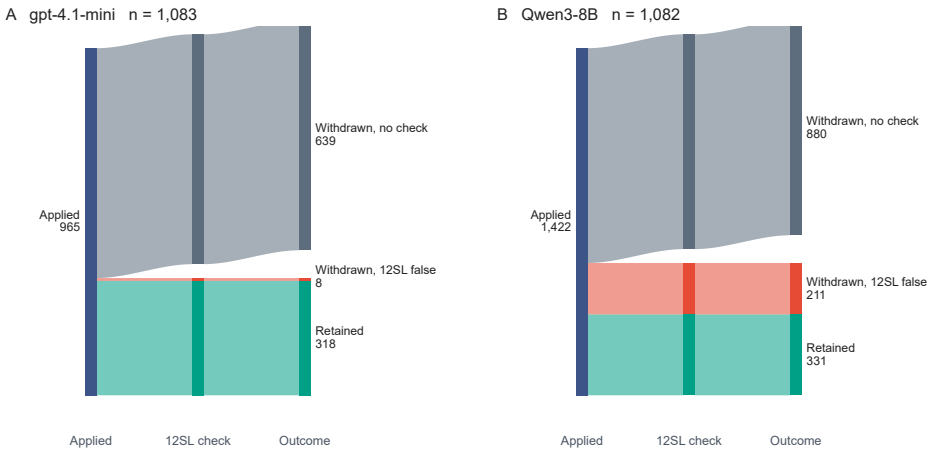


Fig. S6 Applied criteria, the 12SL check, and the withdrawal outcome. Counts are paired notes: 1,083 for gpt-4.1-mini and 1,082 for Qwen3-8B. This figure complements Table 4 of the main paper.

14 Discussion S1. Where the same rule does not reproduce the pattern

This discussion complements the specificity controls in the main paper.

The electroencephalogram check and the sleep check are in the paper because a positive electrocardiogram result is easy to generalize past its evidence. On the resting electroencephalogram, retrieval lowered the case rate and the edit changed nothing. The model did not apply the unsupported spectral cards the rule was ready to withdraw, and it left the true group unstated in about two thirds of notes. On Sleep-EDF, retrieval raised the case rate from 20% to 65%, and the edit left it there. The criterion term was 5% and then zero. The added errors were stage sentences. Restricting the sample to epochs whose true stage already had probability at least one half did not restore the electrocardiogram pattern. The failure is not an excuse provided by a hopeless probability model, although the stage model is hopeless for wake, N1, and REM and solid for N2 and N3.

The common limit is the scope of the edit. The rule rewrites an applied criterion. It does not rewrite a diagnosis, a group label, or a sleep stage. Whenever the harm of retrieval lives in those sentences, the measurement check is silent, and a score that mixes the sentences with the criteria will report that the check failed. Whenever the harm lives in untestable cards or in predicates the measurements reject, the check will report a large drop, and a score that uses the same predicate will agree with it for a definitional reason. Both outcomes occurred in this study. Reporting only the electrocardiogram combined rate would have hidden the second outcome. Reporting the sleep case rate as a failure of measurement checking would have hidden that the criterion term did fall to zero.

Noise and the waveform network bound a different question: whether the predicates themselves move when the tracing is disturbed or when the source archive changes. Recorded baseline, electrode, and muscle noise produce few false releases at modest amplitude and more false absences at the highest muscle mix. A spectral gate is not a general repair. The network's MIMIC transfer is strong for heart rate and QRS duration and weaker in precision for the less common long-PR predicate. None of those experiments rewrote the notes. They support the use of 12SL and Uni-G as the measurement sources for the frozen sample, and they do not extend the note result to a new archive. MIMIC cannot host the eight-check note comparison until the missing measurements exist.

15 Discussion S2. What changes when the library contains only testable cards

This discussion complements Table 6 of the main paper and the controlled retrieval experiments reported there.

The numeric-only generation is the control the full library lacked. With the 33 untestable cards removed before retrieval, the release score falls from 46.8% to 13.8%. Most of the applications that remain unsupported are missing card identifiers: 113 empty and 15 invented. Predicates the algorithm rejected averaged 0.024 per note.

The expert-label gap stays, at 9.8% against 7.5%, and statement mismatch stays near 22% after the agreement rule. A cleaner library removes the artifact that inflated the release score. The stored diagnosis is where retrieval left it.

Two further details keep the control from being over-read. First, the prompts contained 1.32 cards on average, not three, because the numeric library does not always offer three overlaps for a superclass query. The model still applied 0.386 supported criteria per note, more than the 0.294 of the full library, so the shorter prompts were not an empty retrieval. Second, Uni-G rejection was 7.9% before the 12SL edit and 5.5% after it. That is a larger external movement than the 5.3% to 4.6% of the full library, and it is still a small movement next to the release-score change. The numeric cards the closed model chooses to apply are usually acceptable to 12SL. A minority are acceptable to 12SL and not to Uni-G. The agreement rule removes that minority by reading Uni-G. It does not improve the expert-label score.

The practical recommendation that follows is modest. If a system is going to retrieve electrocardiogram criteria into a note, the library should contain predicates, the prompt should require a card identifier from the retrieved set, and a criterion should stay applied only when the measurements accept it. The identifier failures in the numeric-only run, 128 applications that named nothing in the shown set, are a prompt-compliance failure the withdrawal also catches, because a missing check is withdrawn. Catching them is useful. Calling the resulting note more faithful to the stored diagnosis than a note that cited no criterion is not supported by these data.

The matched-instruction generation qualifies the zero release score of the original no-retrieval arm. That zero was partly the prompt, which forbade criterion claims. When both arms may cite a criterion, the arm with no cards has a release score of 17.4%, and every application lacks a check. The keyword arm is still higher, at 31.1%, and its expert-label rate is still higher, 10.6% against 7.6%. The comparison was biased in size. It was not biased in direction.

The schema generation qualifies the 13.8% release score on the numeric library. Of the 128 bad identifiers in the free-JSON run, none remain when the schema requires a retrieved identifier: 469 applied criteria, zero empty, zero invented, and a release score of 3.0%. The 39 withdrawals are all false 12SL predicates, and Uni-G rejects all 39. The residual 5.7% Uni-G record rate is the set 12SL accepted and Uni-G did not. The expert-label rate, 6.0%, is not distinguishable from the original no-retrieval rate of 7.5%. A valid identifier removes a formatting error. The stored diagnosis is a separate question, and the edit still does not answer it.

16 Discussion S3. Additional generations and where the rule does not reproduce

This discussion complements Table 6 of the main paper and the specificity controls. Full details are in Discussion S1 and S2.

The numeric-only generation is the control the full library lacked. With the 33 untestable cards removed, the release score falls from 46.8% to 13.8%. Most of the remaining unsupported applications are missing card identifiers. Rejected predicates are the smaller share. The expert-label gap stays. Statement mismatch stays near

22% after the agreement rule. The rule does not reproduce this pattern outside the electrocardiogram. On the resting electroencephalogram the edit changed nothing, because the model declined to apply the spectral cards. On Sleep-EDF the added errors were stage sentences the rule may not edit. The common limit is the scope of the edit. It rewrites an applied criterion. It does not rewrite a diagnosis, a group label, or a sleep stage.

17 Discussion S4. Why a note with no retrieval can look safer

This discussion complements the exploratory diagnosis rates in Table 7 of the main paper, the different-schema comparison in Table 8, and the micro-F1 changes in Figure 7.

On the prespecified cue list, no retrieval has the lower expert-label rate, 7.5% against 10.1%. That list undercounts affirmative diagnoses. The label-aware rates on the same notes are 19.8% and 23.0%, and other models do not all move in that direction. In the matched pair, the no-retrieval arm retains slightly better overall diagnosis agreement. Micro-F1 is 0.804 against 0.795 (difference −0.009, interval −0.014 to −0.004) and recall is 0.770 against 0.752. Constrained retrieval is the arm that offers measurement-checkable criteria. Hypertrophy recall still falls, from 0.617 to 0.514 (difference −0.103, interval −0.159 to −0.047). The earlier drop to a recall of 0.546 was an unmatched prompt. The withdrawal rule cannot close a diagnosis gap, because it does not edit diagnoses and does not invent statements for codes the model never mentioned.

The no-card schema in the matched pair cannot emit a criterion, so a lower false-diagnosis rate there is not evidence about verified criteria. The comparison that allows criterion claims in both arms raises unsupported criteria from 17.4% to 31.1%. Verified retrieval is the relevant draft when a criterion citation is required and must be one the measurements accept. Treating a single rate as a global ranking mixes that job with diagnosis agreement. Even among applications that survived the agreement rule, only 27.5% of the closed model's linked checks and 12.8% of Qwen's had the linked code in the file. A supported Sokolow–Lyon voltage is not a left-ventricular hypertrophy diagnosis.

No new reader scored the edited sentence, and the prepared review sheet has an empty verdict column; that absence is the main limitation of the study.

18 Waveform-network transfer to 200 MIMIC-IV-ECG studies

This table complements the waveform-network paragraph in the main paper.

A four-head convolutional network was trained on 15,700 recordings and calibrated on 1,972. It reached a calibration macro AUROC of 0.990 and a test macro AUROC of 0.981 on 1,991 recordings for wide QRS, long PR, left axis, and tachycardia. On 200 MIMIC-IV-ECG studies the same heads had AUROC 0.962, 0.957, 0.943, and 0.998, with AUPRC 0.955, 0.592, 0.803, and 0.989 at prevalences 0.125, 0.085, 0.115, and

Table S2 Waveform-network transfer to 200 MIMIC-IV-ECG studies.

Predicate	AUROC	AUPRC	Prevalence
QRS ≥ 120 ms	0.962	0.955	0.125
PR ≥ 200 ms	0.957	0.592	0.085
Axis $\leq -30°$	0.943	0.803	0.115
Heart rate ≥ 100	0.998	0.989	0.145

0.145. This is a transfer of four predicates. It is not a transfer of the note protocol. MIMIC-IV-ECG machine measurements, one recording per subject, can support only part of the library. QRS duration at least 120 ms occurred in 9.57% of 161,116 subjects with a known value, against 7.73% of 1,876 PTB-XL test patients. PR at least 200 ms occurred in 8.01% where the interval was present, against 10.02%. Axis at or below $-30°$ occurred in 11.0%, against 10.0%. Heart rate at least 100 occurred in 12.4%, against 6.56%. Sokolow–Lyon voltage, ST amplitude in II, Q duration, and sex-specific QTc are not in that global file.